

Unit 4.4: Subgroup Performance and Model Reporting

DAIR-3 Workshop, Summer 2026

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Overview

- We have several ways to measure overall model performance:
 - MSE and RMSE,
 - MAE,
 - R^2 ,
 - cross-validated performance,
 - final held-out test performance.
- A regulator, scientist, or program administrator may also ask:
 - Does the model perform differently for different groups?
- How might we report all of these figures to non-technical stakeholders?

Subgroup performance

Subgroup performance

- One aspect of model performance a regulator may care about is subgroup behavior.
- For each subgroup, ask:
 - How many observations are in this group?
 - What is the group's MAE or RMSE?
 - Does the model systematically overpredict or underpredict?
 - Is the pattern large enough to matter in the application?

Mean residuals

- Throughout this notebook:

$$\text{residual} = \text{actual} - \text{predicted}$$

- Positive mean residual means the model **underpredicts** for that group.
- Negative mean residual means the model **overpredicts** for that group.
- **Important:** For a linear model with an intercept term, the mean residual across the full **training set** is **always exactly 0**. So, the average residual is not meaningful for the whole dataset, but rather for a subset.

Discuss: iBudget subgroup performance

- Open:

<https://apd.myflorida.com/resources/reports/APD%20Algorithm%20Report.pdf>

- Find one subgroup or population segment where model performance is discussed. (Don't search for "subgroup" – search for "overestimate".)
- What metrics did they provide? What would you like to have seen?
- In the iBudget example, what are the harms of a model underestimating the true y_i 's for a particular subgroup?

Birthweight subgroup performance

- We'll fit one `Pipeline` on the birthweights dataset together, then inspect its errors by subgroup.
- See if you can understand how the `Pipeline` below is constructed. Can you explain to stakeholders how it works?

In [2]:

```
births = load_birthweight_1971()
births = births[[
    "birthweight", "sex", "momrace", "momage", "dadage", "plurality", "birthorder"
]].dropna().copy()

X = births.drop(columns=["birthweight"])
y = births["birthweight"]

X_birth_train, X_birth_test, y_birth_train, y_birth_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=23,
)
```


In [3]:

```
def add_birthweight_features(X):
    X = X.copy() # Ensures modifications aren't made to the original DataFrame.
    X["multiple_birth"] = np.where(X["plurality"] > 1, "yes", "no")
    X["momage_band"] = pd.cut(
        X["momage"],
        bins=[0, 19, 24, 29, 34, 39, 60],
        labels=["under 20", "20-24", "25-29", "30-34", "35-39", "40+"],
    ).astype("object")
    return X

def birthweight_feature_names(transformer, input_features=None):
    if input_features is None:
        input_features = transformer.feature_names_in_
    return np.array(list(input_features) + ["multiple_birth", "momage_band"], dtype=object)

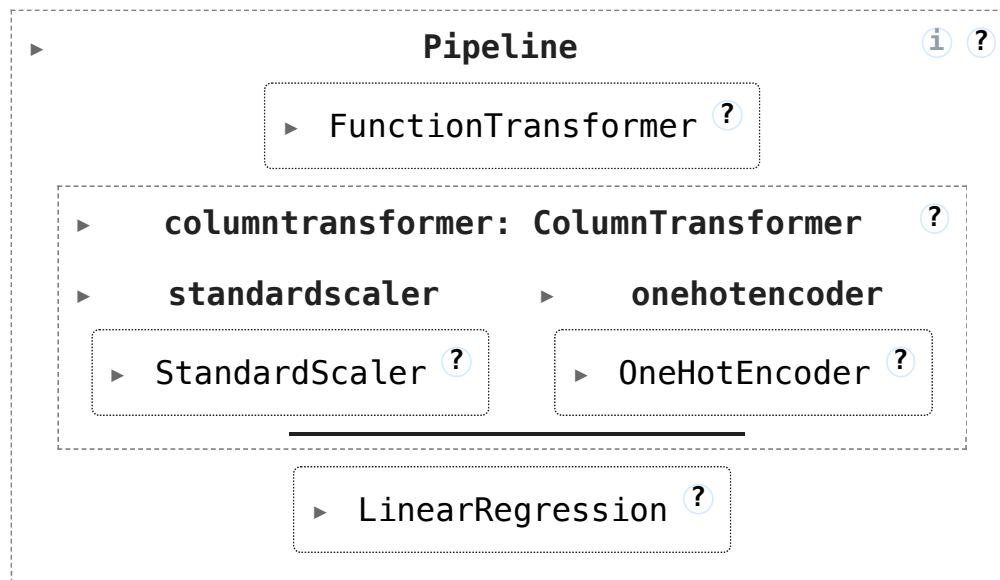
numeric_features = ["momage", "dadage", "birthorder"]
categorical_features = ["sex", "multiple_birth", "momage_band"]

birth_preprocessor = make_column_transformer(
    (StandardScaler(), numeric_features),
    (OneHotEncoder(drop="first", handle_unknown="ignore"), categorical_features),
    remainder="drop",
)

birth_model = make_pipeline(
    FunctionTransformer(
        add_birthweight_features,
        feature_names_out=birthweight_feature_names,
    ),
    birth_preprocessor,
    LinearRegression(),
)

birth_model.fit(X_birth_train, y_birth_train)
```

Out [3]:



In [4]:

```
# Make sure you understand why there are 10 numbers here!
birth_model[-1].coef_
```

Out[4]:

```
array([ 60.81147644, -4.62147396,  6.23290764, 124.00163727,
       -945.61256996, -17.07714555, -51.78106843, -108.75499514,
       -161.19704101, -42.28378678])
```

In [5]:

```
# model_diagnostics is a helper function defined in setup.py
birth_diagnostics = model_diagnostics(
    birth_model,
    X_birth_train,
    X_birth_test,
    y_birth_train,
    y_birth_test,
)
birth_diagnostics
```

Out [5]:

	train	test
MSE	315285.642339	315455.342682
RMSE	561.503021	561.654113
MAE	420.084938	420.253557
R^2	0.064344	0.063480

Activity: Overall performance

- Before we look at subgroups, write one plain-English sentence summarizing the overall model performance, like you might have seen in the iBudget report.

In []:

Computing residuals

- The code below computes residuals on the **test set**. Remember, the test set is our best guess of what "future data" may look like.

In [6]:

```
test_predictions = birth_model.predict(X_birth_test)
report_frame = X_birth_test.copy().assign(
    actual=y_birth_test.to_numpy(),
    predicted=test_predictions,
)
report_frame["residual"] = report_frame["actual"] - report_frame["predicted"]
report_frame.head()
```

Out[6]:

	sex	momrace	momage	dadage	plurality	birthorder	actual	predicted	residual
1037139	male	White	20	22.0	1	1.0	3345.0	3352.008782	-7.008782
181863	male	White	23	23.0	1	2.0	1503.0	3388.595705	-1885.595705
1701201	female	White	26	26.0	1	4.0	3232.0	3286.002960	-54.002960
91949	female	White	38	36.0	1	2.0	3941.0	3316.570102	624.429898
738186	male	White	44	47.0	1	7.0	2920.0	3464.882884	-544.882884

- The helper function below computes various measures of model performance, again on the test set.

In [7]:

```
subgroup_metrics(report_frame, group_col="sex")
```

Out[7]:

	n	MSE	RMSE	MAE	R^2	mean_residual
sex						
male	163120	334511.657099	578.369827	432.224306	0.051378	-2.016469
female	155137	295418.433320	543.524087	407.666821	0.055597	2.449080

In [8]:

```
# To give you a sense of where the numbers above are coming from.  
report_frame.loc[report_frame['sex'] == 'male', 'residual'].abs().mean()
```

Out[8]:

432.22430610419264

Discuss: Subgroup performance by maternal race

- Which group has the largest MAE? Smallest?
- Which group has the largest mean residual? Smallest?
- Are the differences large enough to mention in a report?
- Which groups have small enough sample sizes that you would be cautious?

In [9]:

```
subgroup_metrics(report_frame, group_col="momrace")
```

Out [9]:

	n	MSE	RMSE	MAE	R^2	mean_residual
momrace						
Black	37276	384908.398739	620.409864	451.991696	-0.019833	-180.748922
Native	1721	319456.855128	565.205144	424.232870	0.026538	41.678865
White	276699	306333.692740	553.474202	416.140028	0.060312	25.828014
Asian	2561	287393.139137	536.090607	400.062251	-0.075553	-167.786719

Discuss: Subgroup performance by plurality

- How much worse is performance for twins or triplet-plus births?
- Is the model overpredicting or underpredicting for any group?
- Why might this happen, given the features in the model?

In [10]:

```
report_frame
```


Out [10]:

	sex	momrace	momage	dadage	plurality	birthorder	actual	predicted	residual
1037139	male	White	20	22.0	1	1.0	3345.0	3352.008782	-7.008782
181863	male	White	23	23.0	1	2.0	1503.0	3388.595705	-1885.595705
1701201	female	White	26	26.0	1	4.0	3232.0	3286.002960	-54.002960
91949	female	White	38	36.0	1	2.0	3941.0	3316.570102	624.429898
738186	male	White	44	47.0	1	7.0	2920.0	3464.882884	-544.882884
...	...	...	...	...	...	...	...	...	...
743758	male	White	32	35.0	1	6.0	4167.0	3443.532518	723.467482
833627	female	Black	32	27.0	1	5.0	3430.0	3321.867767	108.132233
730122	male	White	26	28.0	1	2.0	3033.0	3401.970364	-368.970364
172577	female	White	34	25.0	1	6.0	4139.0	3349.245422	789.754578
623866	male	White	24	35.0	1	1.0	3119.0	3388.139853	-269.139853

318257 rows × 9 columns

In [11]:

```
subgroup_metrics(report_frame, group_col="plurality").sort_index()
```

Out[11]:

	n	MSE	RMSE	MAE	R^2	mean_residual
plurality						
1	312534	313088.731922	559.543324	418.547142	0.018408	0.188127
2	5632	440791.053529	663.920969	510.878646	0.019507	6.605759
3	90	693530.953957	832.785059	677.166150	-0.554387	-502.151916
4	1	44254.086612	210.366553	210.366553	NaN	210.366553

Activity: Document the model's performance

- Write 1-2 sentences describing the model's overall performance, including its likelihood of generalizing to unseen data.
- State the number of parameters used in the model, and name the parameter with the largest magnitude.
- Write 2-3 sentences describing its performance on various subgroups.
- Ensure that what you write is

Submit your work to ALICE. **Use the prompt here, rather than what is written in ALICE, as your assessment for Unit 4.4.**

In [12]:

```
birth_model[:, -1].get_feature_names_out()
```

Out[12]:

```
array(['standardscaler__momage', 'standardscaler__dadage',  
      'standardscaler__birthorder', 'onehotencoder__sex_male',  
      'onehotencoder__multiple_birth_yes',  
      'onehotencoder__momage_band_25-29',  
      'onehotencoder__momage_band_30-34',  
      'onehotencoder__momage_band_35-39',  
      'onehotencoder__momage_band_40+',  
      'onehotencoder__momage_band_under 20'], dtype=object)
```

In []:

Key takeaways

- Overall performance can hide uneven subgroup performance.
- MAE and RMSE describe error magnitude; mean residual describes direction.
- Positive mean residual means underprediction; negative mean residual means overprediction.
- Subgroup metrics should always be read with group sizes.
- A model report should state what the model can and cannot safely be used for.

Thanks!

- A few of you have asked for materials from my courses, which dive deeper into some of these topics:
 - For more materials on practical machine learning: **practicaldsc.org**.
 - Linear algebra and machine learning: **eeecs245.org**.
- Feel free to reach out: rampure@umich.edu